



Pathfinder

Resume Matcher & Skill
Recommender

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What This App Does

- Analyses resume vs job description
- Extracts skills using keyword matching
- Calculates match percentage
- Identifies missing skills
- Suggests Coursera & Udemy courses
- Generates a PDF report
- Saves and displays user history
- Includes login & registration system



Authentication System



Login & Registration
via Streamlit UI



Credentials stored in
users.json



Each user has profile
image, history, and
data stored



Logout & Clear
History options in
sidebar

Skill Extraction Logic



Predefined list of skill keywords



Extracts text from PDF or DOCX



Searches for keyword patterns using regex



Produces Resume Skills & Job Skills

Matching Mechanism

Intersects Resume & Job skills



Identifies matched and missing skills



Calculates match % = $\frac{\text{matched}}{\text{job skills}}$



Displays bar graphs & pie charts using Plotly

Course Recommendations



Dataset lookup from Coursera & Udemy CSV files



Finds first matching course for each missing skill



Generates links for user convenience

PDF Report Generation

- Uses ReportLab
- Includes match %, skills, missing skills
- Lists recommended courses with URLs
- Automatically downloadable

User History Tracking



Saves each session's match result



Shows past timestamps & matches in sidebar



Stores file name of generated PDF



Can clear entire history

Overall Workflow

